Bowl-shaped and particle-in-hole gold nanopores at 785 nm: open-source FDTD simulations

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Summary

Both geometries suggested by Dr. Huang were simulated with Meep (open-source FDTD, MIT). The bowl-shaped nanopore of Zhao *et al.*, Nano Lett. 23, 4830 (2023) was modelled from Table S1/Fig. S2 of its Supporting Information; because the structure is axisymmetric it is solved exactly in cylindrical coordinates, which allows 0.5-1 nm grids.

• **Validation.** For a 50 nm Au sphere in water the solver agrees with exact Mie theory within 1.8 % at 5 nm and 0.9 % at 10 nm from the surface (0.5 nm grid, 780 nm), and the error decreases with mesh refinement.
• **Conventional pore.** Where the geometry is fully specified the result matches Fig. S4: R = 2.5 nm resonates at 640 nm (paper 640 nm) with $|E|^2/|E_0|^2$ of about 12 (paper ~10).
• **Bowl pore.** One hot spot at the pore rim, as in Fig. 1D-F. The resonance is set mainly by the Au thickness at the rim h_3: 650 nm for h_3 = 10 nm to 800 nm for 2.5 nm. With h_3 = 3 nm (within the 2.5-10 nm range of Table S1), a 1 nm rounded rim and the 500 nm Gaussian beam of SI Note 2, the resonances are 790 nm (R = 10 nm), 710 nm (R = 2.5 nm) and 700 nm (R = 1 nm), against 785, 710 and 640 nm in Fig. S4.
• **Hot-spot size.** Along the pore wall (the direction in which DNA translocates) the intensity stays above half maximum over 3.3-3.9 nm, and it falls to 20-30 % within 2 nm of the metal, consistent with the 2.4-3.1 nm hot spot and the 2 nm decay reported in the paper.
• **Absolute enhancement.** The resonance position is mesh-converged, the peak $|E|^2$ at the rim is not: it keeps rising with refinement because the rim is a (near-)sharp metal edge. Absolute values such as the ~140 of Fig. S4 therefore depend on the edge radius and the mesh, and I report them with that caveat.
• **Particle in the 200 nm nanohole.** A 50 nm Au sphere near the wall creates a gap hot spot along the polarisation, as in Fig. 1e of the Nat. Commun. paper; 3D values are given in Section 6 with their mesh dependence.

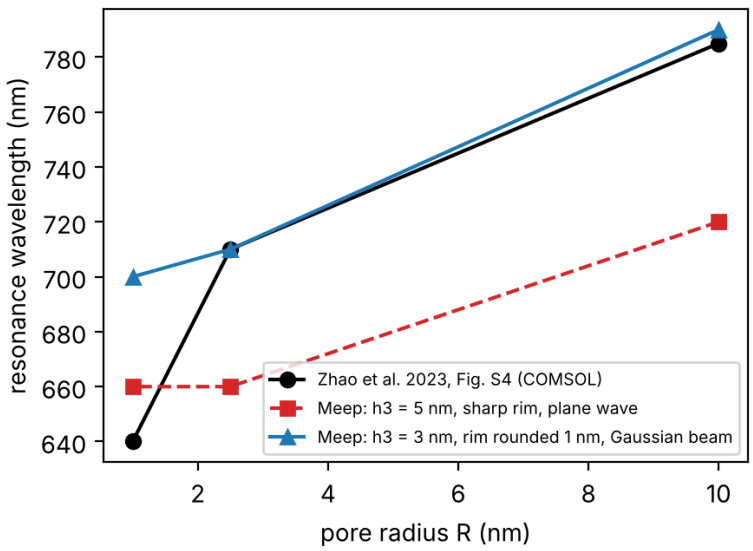


Figure 1. Resonance wavelength of the bowl nanopore versus pore radius: Fig. S4 of the paper (black) and Meep, for the nominal h_3 = 5 nm with a sharp rim (red) and for h_3 = 3 nm with a 1 nm rounded rim and the 500 nm Gaussian beam of SI Note 2 (blue). 0.67 nm grid.

1. Method

Geometry (Table S1, Fig. S2). Au layer h_1 = 100 nm on Si_3N_4 h_2 = 30 nm; bowl opening D = 250 nm; Au thickness at the pore h_3; pore radius R through the Au and the SiN. The bowl surface is the sphere through (D/2, h_1) and (R, h_3) centred on the axis. Water n = 1.33, SiN n = 2.0 (the SI lists the two values in swapped order). Excitation along -z from the Au side, 550-1000 nm in one broadband run.

Gold. Rakic *et al.*, Appl. Opt. 37, 5271 (1998), the reference of the SI. The 6-pole Lorentz-Drude model distributed with Meep diverges on sub-nanometre grids when the Au rim is 5 nm thick or thinner, so the same Rakic permittivity was refitted between 500 and 1050 nm with one Drude and two Lorentz poles (maximum relative error 0.19 %, ε(785 nm) = -19.26 + 2.03i). A Johnson-Christy run is included for comparison.

Axisymmetric solution. For normal incidence on an axisymmetric structure a linearly polarised beam is the sum of the m = +1 and m = -1 angular orders, which are mirror images of each other. One 2D (r, z) run with m = +1 and a source E_r = 1, E_φ = i gives the full 3D field: $|E|^2(r, \varphi, z) = (|E_r|^2 + |E_z|^2)\cos^2\varphi + |E_\varphi|^2 \sin^2\varphi$. This makes 0.5 nm grids affordable.

Normalisation. $|E_0|^2$ is the incident intensity at the pore position from an identical run without the structure (water only), on the same grid and with the same source (plane wave or Gaussian).

Rim shape. Runs marked 'rounded' replace the two 90° Au corners of the rim (top and bottom of the h_3 edge) by quarter circles of 1 nm radius.

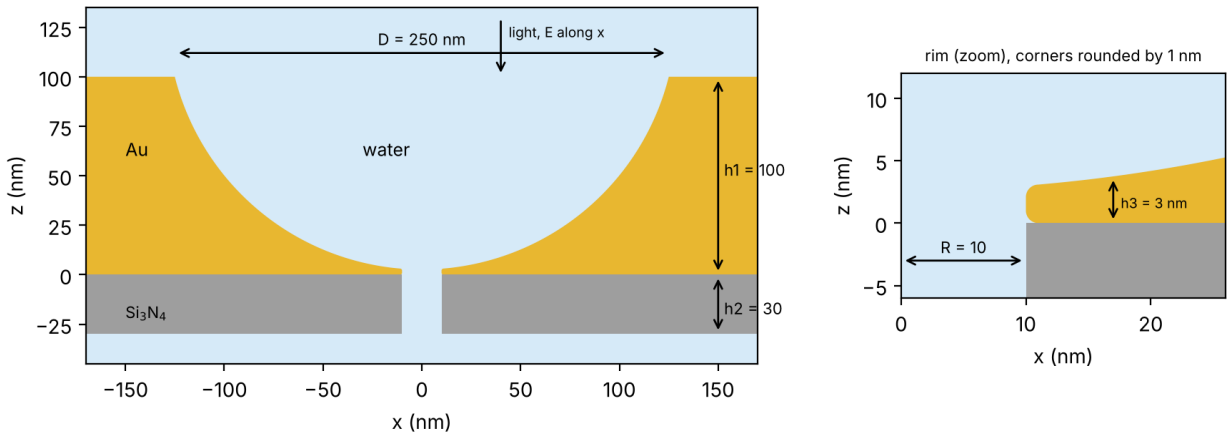


Figure 2. Modelled cross-section (R = 10 nm, h_3 = 3 nm) with the dimensions of Table S1; right: the rim with its two corners rounded by 1 nm. The structure is rotationally symmetric about the pore axis (x = 0).

2. Validation against Mie theory

grid	1 nm	2 nm	5 nm	10 nm (distance from surface)
1 nm	-10.7 %	-7.5 %	-4.2 %	-2.9 %
0.5 nm	-8.3 %	-4.9 %	-1.8 %	-0.9 %

Table 1. Relative error of $|E|^2/|E_0|^2$ along the polarisation for a 50 nm Au sphere in water at 780 nm, Meep versus exact Mie theory (scattnlay), same Rakic permittivity.

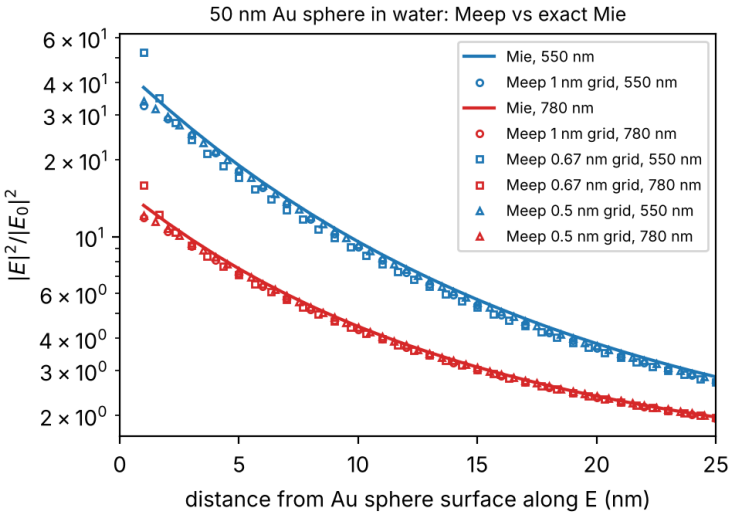


Figure 3. Near field of a 50 nm Au sphere in water: Meep on 1, 0.67 and 0.5 nm grids (symbols) and exact Mie theory (lines), at 550 and 780 nm.

3. Bowl-shaped versus conventional nanopore

case	grid	λ_res (nm)	paper λ_res	paper max
conventional R = 10 nm, Au 5 nm	1 nm	880	840	~40
conventional R = 2.5 nm, Au 5 nm	0.67 nm	640	640	~10
bowl R = 10, h3 = 5, sharp rim	1 / 0.67 / 0.5 nm	710 / 720 / 720	785	~140
bowl R = 10, h3 = 5, rounded, Gaussian	1 / 0.67 nm	710 / 720	785	~140
bowl R = 10, h3 = 3, rounded, Gaussian	1 / 0.67 nm	770 / 790	785	~140
bowl R = 2.5, h3 = 3, rounded, Gaussian	0.67 nm	710	710	~250
bowl R = 1, h3 = 3, rounded, Gaussian	0.67 nm	700	640	~500

Table 2. Resonance wavelengths (maximum $|E|^2$ in the water next to the rim). Paper values are read from Fig. S4.

The conventional pore reproduces Fig. S4 (exactly for R = 2.5 nm; 880-890 nm against 840 nm for R = 10 nm). The bowl pore concentrates the field in a single hot spot on the few-nanometre metal edge of the rim (Fig. 4, 5), where the bowl funnels the incident light, and is resonant anywhere in the 650-800 nm range depending on the rim, as discussed in Section 5.

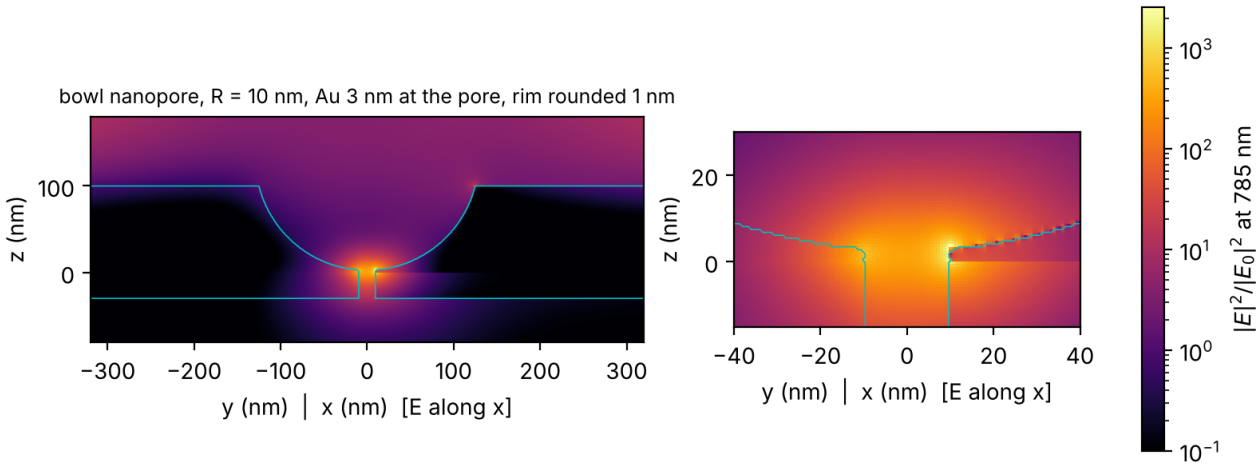


Figure 4. $|E|^2/|E_0|^2$ at 785 nm (log scale), cross-section of the bowl pore; left half in the plane perpendicular to E, right half in the plane of E. Right: zoom on the pore.

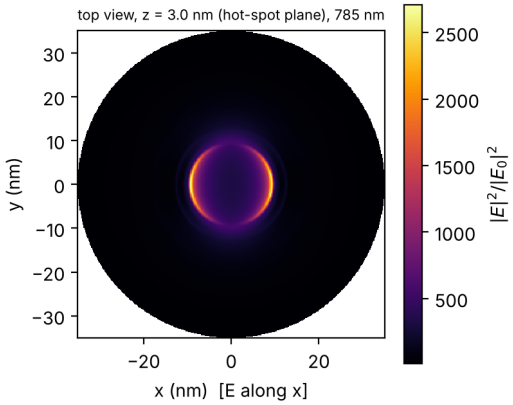


Figure 5. Top view in the plane of the hot spot, 785 nm: two lobes on the rim along the polarisation, as in Fig. 1E of the paper.

4. Size of the hot spot

For single-molecule SERS the size of the probed volume matters as much as the enhancement. The paper infers a hot spot of 2.4-3.1 nm (7-9 consecutive bases) from the Raman time traces and states that the plasmonic field drops significantly 2 nm from the metal surface. Figure 6 shows the intensity at 785 nm along two lines through the hot spot: from the Au wall into the pore, and along the pore wall, which is the direction of DNA translocation.

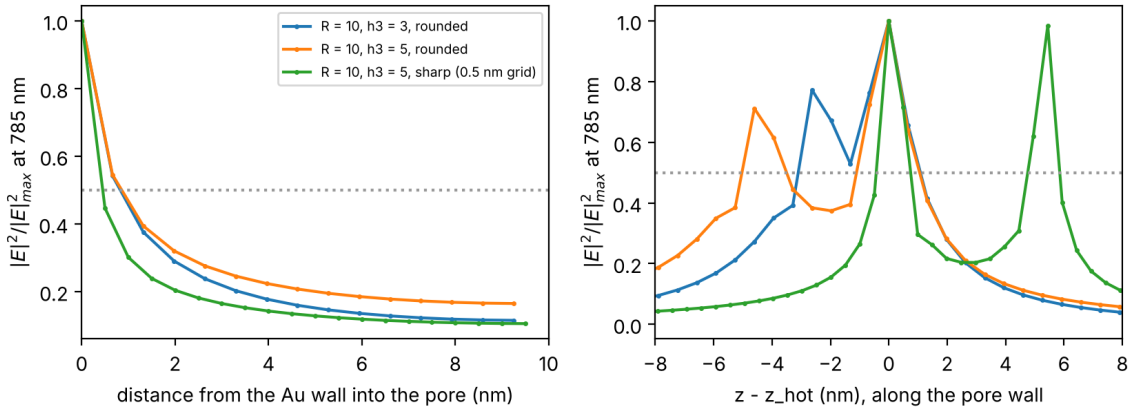


Figure 6. $|E|^2$ normalised to its maximum at 785 nm, R = 10 nm. Left: from the Au wall towards the pore axis at the height of the hot spot. Right: along the pore wall (z), just inside the pore; the two maxima are the upper and lower edges of the Au rim.

case	grid	length > half max	at 2 nm from Au	azimuthal FWHM
bowl R = 10, h3 = 3, rounded	0.67 nm	3.9 nm	0.29	104°
bowl R = 10, h3 = 5, rounded	0.67 nm	3.3 nm	0.32	108°
bowl R = 10, h3 = 5, sharp	0.5 nm	2.0 nm	0.20	100°
Zhao et al. 2023 (experiment)		2.4-3.1 nm	drops within 2 nm	

Table 3. Extent of the rim hot spot at 785 nm. 'Length > half max' is the total length of the pore wall where $|E|^2$ exceeds half its maximum; 'at 2 nm from Au' is $|E|^2/|E|^2_{max}$ 2 nm from the wall towards the pore axis; the azimuthal width is that of each of the two lobes along the polarisation.

Along the pore wall the hot spot covers the thickness of the Au rim, with maxima at its upper and lower edges, and stays above half maximum over 3.3-3.9 nm, the same scale as the 2.4-3.1 nm inferred from the Raman data. Away from the metal the intensity halves within the first nanometre (resolved by only one or two grid cells) and is down to 20-30 % at 2 nm, in line with the 2 nm decay discussed in the paper. Azimuthally the hot spot forms two lobes along the polarisation, each about 100° wide (~17 nm of rim for R = 10 nm, Fig. 5). For R = 2.5 and 1 nm the intensity does not fall below half maximum anywhere inside the pore: the hot spot fills the whole pore, the 'no cut-off' behaviour the paper attributes to the bowl geometry.

5. What sets the resonance and the enhancement

Rim thickness. Thinning the Au at the rim from 10 to 2.5 nm moves the resonance from 650 to 800 nm (Fig. 7), about 12-36 nm per nanometre of Au, most steeply for the thinnest rims. Table S1 gives h_3 as 2.5-10 nm and Fig. S2 as 5-10 nm, so the 785 nm of Fig. S4 lies inside the range spanned by the published geometry; with h_3 = 3 nm the model resonates at 790 nm. Details not given in the paper (exact bowl profile near the pore, the 2 nm Ti adhesion layer, the 5-10 nm Au overcoat on the pore wall) act on the same thin rim and are expected to shift the resonance by similar amounts.

Pore radius. With the same rim, smaller pores resonate at shorter wavelength, as in Fig. S4 (Fig. 1). A 40 nm pore with h_3 = 5 nm resonates at 750 nm.

Absolute $|E|^2$. The hot spot sits on the edge of the rim. For a sharp 90° edge the peak field grows without limit as the mesh is refined (R = 10 nm, h_3 = 5 nm: 1933, 2965 and 3803 on 1, 0.67 and 0.5 nm grids), while the resonance wavelength stays at 720 nm. Rounding the edge by 1 nm lowers the peak, but a 1 nm radius is itself only resolved on grids well below 0.5 nm. Converged absolute enhancements therefore require the actual edge radius of the fabricated rim and a mesh finer than that radius; the value reported in Fig. S4 depends on the same choices in the COMSOL model.

case	1 nm	0.67 nm	0.5 nm
sharp rim, h3 = 5: resonance (nm)	710	720	720
sharp rim, h3 = 5: peak $\|E\|^2/\|E0\|^2$	1933	2965	3803
rounded, h3 = 5: resonance (nm)	710	720	-
rounded, h3 = 5: peak $\|E\|^2/\|E0\|^2$	432	1016	-
rounded, h3 = 3: resonance (nm)	770	790	-
Johnson-Christy, sharp, h3 = 5: resonance (nm)	690	690	-

Table 4. Grid convergence for the bowl pore, R = 10 nm: the resonance wavelength changes by at most one 10 nm spectral step (20 nm for the thinnest rim) under refinement, whereas the peak intensity at the rim keeps growing.

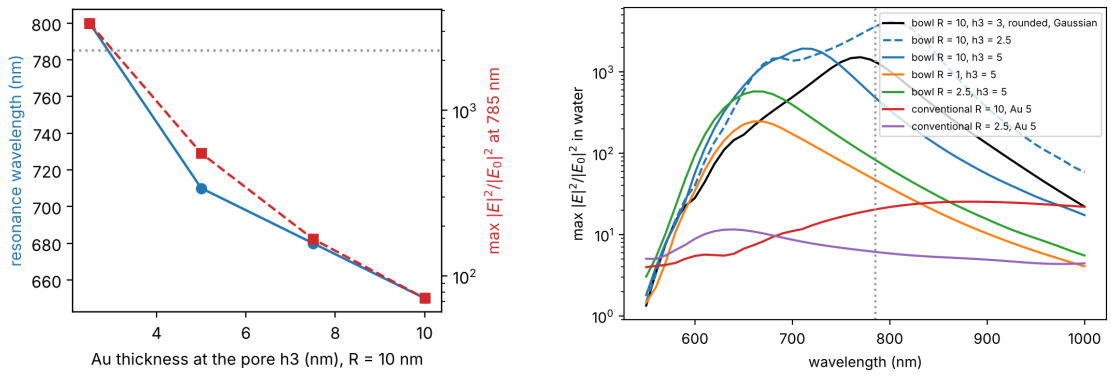


Figure 7 (left). Bowl pore, R = 10 nm, sharp rim, plane wave, 1 nm grid: resonance wavelength (blue) and $|E|^2$ at 785 nm (red) versus the Au thickness at the rim h_3. Figure 8 (right). Maximum $|E|^2/|E_0|^2$ next to the rim versus wavelength for bowl and conventional pores; absolute values are mesh dependent, the peak positions are converged.

6. Particle in the 200 nm Au nanohole (Nat. Commun. 2019)

Full 3D model at 785 nm: 200 nm hole through 100 nm Au on 100 nm SiN, water, plane wave from the Au side polarised along x; one period of a 400 nm hole array (the device is an array; the periodic cell also keeps the incident wave clean). Gold is a single Drude pole fitted exactly to the Rakic permittivity at 785 nm, which gives the exact frequency-domain solution at that wavelength. The nanourchin is replaced by a smooth 50 nm sphere at mid-height of the Au, with its surface 5 or 10 nm from the wall along the polarisation.

case	2 nm grid	1.5 nm grid
hole only: maximum (upper rim)	25	23
sphere, 10 nm gap: maximum	28	24
sphere, 10 nm gap: gap centre	10	5
sphere, 5 nm gap: maximum	74	41
sphere, 5 nm gap: gap centre	68	14

Table 5. $|E|^2/|E_0|^2$ at 785 nm in the water.

The sphere couples to the hole wall and creates a hot spot in the particle-wall gap along the polarisation, as in Fig. 1e of the paper. For a 5 nm gap it is the strongest field in the hole; for a 10 nm gap it is comparable to the hole rim. The hole-rim value is stable between the two grids; the gap values are not yet converged (5-7 cells across a 10 nm gap, 2-3 across 5 nm). A uniform 3D grid fine enough for few-nanometre gaps or urchin tips exceeds the computing time available to me (one 1.5 nm run takes 3 h on 4 cores); a locally refined mesh (FEM, or FDTD with mesh override) would be needed.

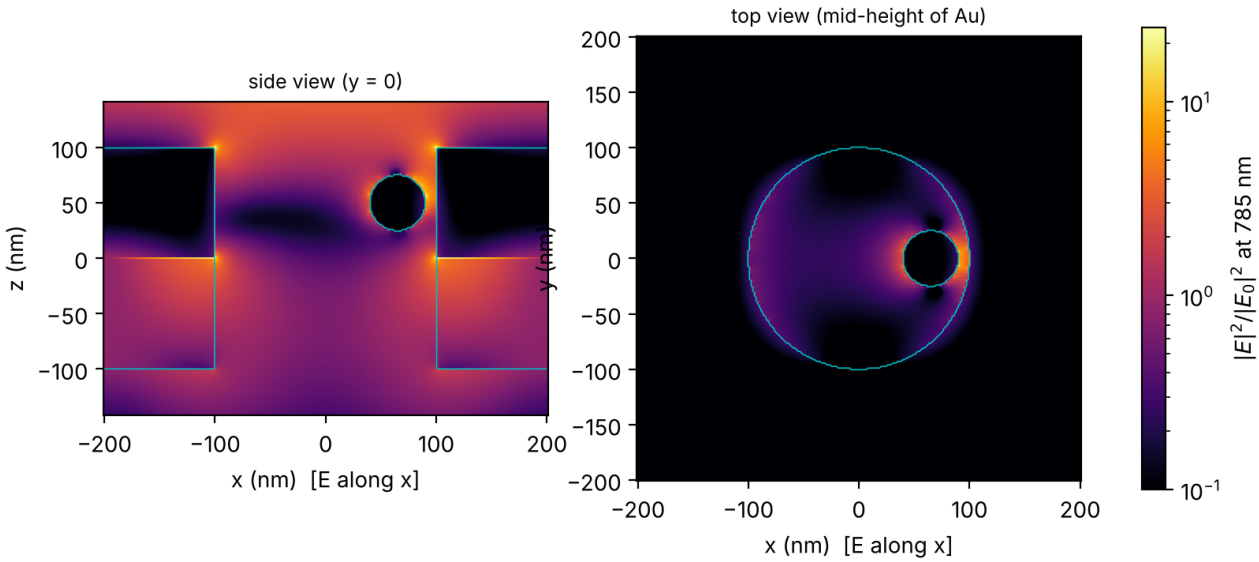


Figure 9. 50 nm Au sphere 10 nm from the wall of a 200 nm Au nanohole, 785 nm, 1.5 nm grid: side view in the plane of E (left) and top view at mid-height of the Au (right).

7. Summary and next steps

The open-source workflow reproduces exact Mie theory and the conventional-pore results of Fig. S4, and the bowl-pore resonances of Fig. S4 within the range allowed by the published geometry. The remaining differences come from the rim geometry (thickness, edge radius, Ti and Au overcoat) and, for absolute enhancements, from the mesh at the rim.

To make the comparison quantitative I would need the COMSOL geometry (or the exact rim profile and thickness), the mesh size at the rim and the illumination side. With these, the natural next steps are (i) converged enhancement maps of the bowl pore for your current pore sizes and laser, (ii) the clustering states of Au particles in the pore from your Raman-feedback work, and (iii) Purcell/LDOS maps for the SERS emission inside the hot spot, which is the subject of my current work with Dr. Bagheri Harouni.

References

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Appendix: numerical notes

• Cylindrical runs: 2D (r, z) cell of 1.25 x 0.88 µm (Gaussian beam) or 0.75 x 0.88 µm (plane wave) with 0.25 µm PML on all sides; broadband pulse, DFT fields recorded until they decay to 10^{-6}.
• The PML crossing the Au film is stable with the refitted gold; with the 6-pole Rakic model of the Meep library the fields diverge (NaN) on ≤ 1 nm grids for $h_3 \leq 5$ nm, and halving the Courant number does not cure it.
• 3D runs: periodic 400 nm cell, PML along z only, mirror symmetry y → -y.
• All runs were executed in parallel on GitHub Actions (4 cores per case); scripts, data and figures are available on request.